NEW SINGLE-FAMILY DWELLING

Drawings for engineering review and building permit preparation - Alberta

PROJECT DATA

Main floor, heated (to wall centre lines)	5,680 ft2	527.6 m2
Attached garage	833 ft2	77.4 m2
Porches and deck (open)	2,028 ft2	188.4 m2
Basement, inside the ICF walls	3,321 ft2	308.5 m2

BUILDING AREA - THE FIRST DECISION

Building area (NBC: the greatest area above grade, to the outside face of the walls) of the house with its attached garage: about 605 m2 at the wall centre lines and 626 m2 to the outside face.

NBC(AE) 2023 Part 9 covers houses of up to 600 m2 building area (Div. A 1.3.3.3). This design is over the limit, so Part 3 would apply as drawn.

Without the garage the house is about 546 m2. Options: detach the garage (a breezeway); separate it with a firewall; trim about 35 m2 (375 ft2) of enclosed area; or design the house to Part 3 with a professional. Confirm with the permit office and decide before the engineer starts: it changes the code path for the whole design.

CODE BASIS

- National Building Code - 2023 Alberta Edition, in force May 1, 2024: Part 9 if the building area is 600 m2 or less (above).
- Energy efficiency: NBC Section 9.36, Tier 1 minimum. The performance path (an energy advisor's HOT2000 model) suits this house best.
- Electrical: Canadian Electrical Code with Alberta modifications. Gas: CSA B149.1. Plumbing: National Plumbing Code. Private sewage: Alberta Private Sewage Systems Standard of Practice 2021.
- Builder: a licensed residential builder, or an Owner Builder Authorization from the Alberta registrar, before the building permit is issued (New Home Buyer Protection Act).

OCCUPANCY

- Group C, one dwelling unit: one storey above grade with a full basement and an attached garage.
- Smoke and CO alarms, the garage separation and the foam thermal barrier: A1.1, A1.2, A5.0.

ABBREVIATIONS

- C/L centre line; EL. elevation (main floor = 0'-0"); T/O top of; U/S underside of; FTG footing; ICF insulated concrete form; P.T. pressure-treated; E.W. each way; O.C. on centre; U.N.O. unless noted otherwise; AHJ authority having jurisdiction; PSDS private sewage disposal system; HRV heat recovery ventilator; NBC(AE) National Building Code - 2023 Alberta Edition.

SYMBOLS

D101 door mark (A6.0)

W101 window mark (A6.0)

A / A4.2 section: letter / sheet

elevation tag

PHASING

Phase 1 - the house, and all of the basement concrete

- Footings, ICF walls and the basement slab.
- The pool and spa shells, with their piping, drains and the hydrostatic relief valve.
- The theater's depressed (stepped) floor.
- The exit stair areaway and stair, the light wells and their guards.
- Frost walls, the garage slab and the porch piers.
- Under-slab plumbing for every future basement fixture, floor drains, the sump and the radon pipe.
- The main stair, the basement bearing walls, beams and posts (S1.1), and the mechanical room (furnace or boiler, HRV, water heater, water treatment, panel).
- The main floor, roof and envelope, finished for occupancy.
- A thermal barrier (1/2" gypsum) over all the ICF foam inside, smoke and CO alarms, and the basement lighting and outlets the code needs.

Phase 2 - finishing the basement, after the house is occupied

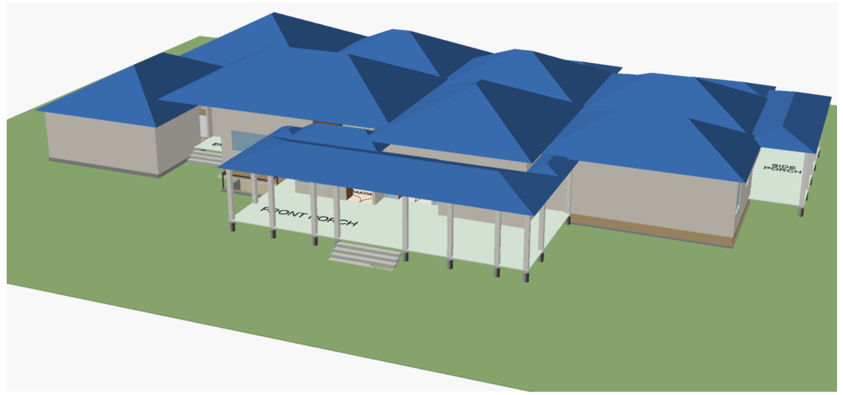
- Partitions, drywall, ceilings and floors; the bathrooms and laundry; the theater (room within a room, seating, screen) and the music room; the pool room finishes.
- Phase 2 takes its own building permit, plus electrical and plumbing permits.

Why the split works

- Everything structural, everything under the slab and everything in the concrete is in Phase 1, so Phase 2 is interior work only - no cutting of the slab, no new footings.

ALBERTA ROADMAP (details: README.md)

- 1. Land: title, land-use district and setbacks; a development permit from the municipality.
- 2. Survey (plot plan) and a geotechnical investigation on the real site.
- 3. Builder: a licensed builder, or an Owner Builder Authorization, plus warranty or its exemption.
- 4. Design: this set reviewed and engineered (APEGA), truss layouts, the 9.36 energy compliance.
- 5. Building permit (municipality or its accredited agency); electrical, gas and plumbing permits; PSDS permit by a certified installer; water well by a licensed driller; road approach.
- 6. Utility locates before digging; inspections at the footings, foundation, framing, insulation and final; occupancy.
- 7. Phase 2: a separate permit set for the basement finish.



DRAWING INDEX

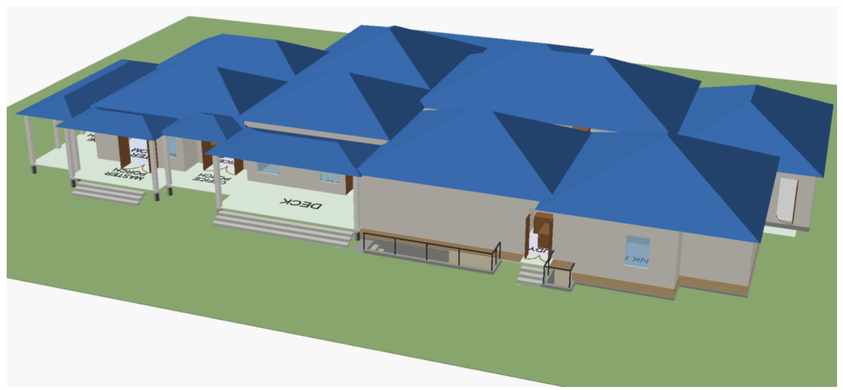
A0.0	COVER, PROJECT DATA, DRAWING INDEX
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S0.1	STRUCTURAL DESIGN BRIEF
S1.0	FOUNDATION PLAN
S1.1	MAIN FLOOR FRAMING CONCEPT
A1.1	BASEMENT PLAN
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A1.3	ROOF PLAN
A3.0	ELEVATIONS
A4.0	BUILDING SECTIONS A AND B
A4.1	BUILDING SECTIONS C, D AND E
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A5.1	DETAILS - POOL, THEATER, STAIRS, LIGHT WELL
A6.0	SCHEDULES - DOORS, WINDOWS, ROOMS

ALSO IN THE PACKAGE (docs/permit/)

- README.md - the engineering brief and the Alberta permit roadmap and checklist, with sources.
- cad/ - the plans as DXF files (inches, layered). schedules/ - door, window and room schedules (CSV).

FLAGS TO RESOLVE BEFORE THE PERMIT

- Building area over the Part 9 limit (left).
- Geotechnical report: bearing, groundwater, sulphates - the pool pit is 18'-3" below grade.
- Porch floors 2'-0" above grade: guards, or finished grade raised within 600 mm.
- Fireplace type (gas or wood) in the great room - it decides the code path for it.
- A surveyed plot plan on the real parcel (A0.1).



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Rural Alberta - legal land description: TBD

Owner / applicant: _____

NOT FOR CONSTRUCTION

FOR ENGINEERING REVIEW AND PERMIT PREPARATION

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ENGINEER'S REVIEW / SEAL

SHEET TITLE

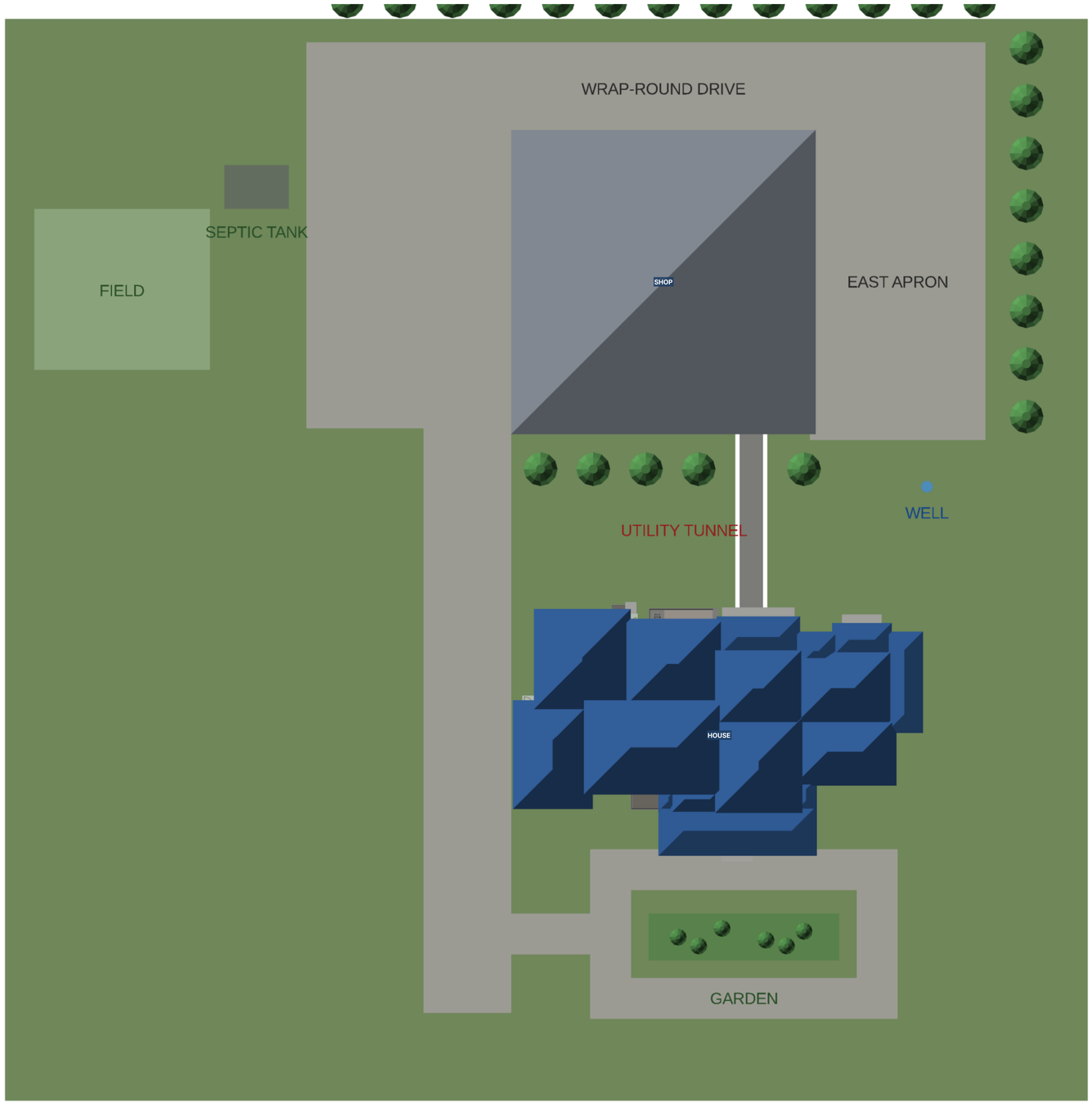
COVER, PROJECT DATA, DRAWING INDEX

SHEET 1 of 14

A0.0

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1 / A0.1 **SITE PLAN - SCHEMATIC, NOT A SURVEY**

SCALE: 1" = 20'-0"

N

0' 20' 40' 80' 160'

THIS IS NOT THE PERMIT SITE PLAN

- Land is not chosen yet: positions are relative to the house, from models/site.scad.
- For the development and building permits, replace this sheet with a plot plan by an Alberta Land Surveyor (ALS) on the real parcel, showing:
- property lines, the legal land description and the parcel area;
- setbacks from every property line and from the road (municipal land-use bylaw);
- easements, rights-of-way, pipelines and power lines (Utility Safety Partners locate);
- the house, the shop and every other building, with their distances apart;
- the water well and the private sewage (PSDS) system with their separation distances (Alberta Private Sewage Systems Standard of Practice 2021);
- the driveway and its approach to the road (municipal approach permit);
- existing and finished grades, the geodetic elevation of the main floor (0'-0" here), and drainage away from the buildings;
- any watercourse, wetland, slope or flood hazard.

SITE SERVICES (separate permits)

- Water well by a licensed driller (household use under 1 250 m3/yr needs no licence; register the well).
- PSDS by a certified installer after a site evaluation (soil test pits).
- Power, gas or propane, and the shop tunnel (to the basement UTILITY room later).

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SITE PLAN (SCHEMATIC)

SCALE: 1" = 20'-0"

SHEET 2 of 14

A0.1

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1. DESIGN CRITERIA (TO BE COMPLETED)

From NBC(AE) 2023 Div. B Appendix C for the site, and the geotechnical report

Location / nearest listed place _____

Climate zone (7A / 7B) and heating degree-days _____

Ground snow load Ss / rain load Sr (kPa) _____

Hourly wind pressure q 1/50 (kPa) _____

Seismic Sa(0.2) / site class _____

Frost penetration (min. footing depth 1.2 m) _____

Bearing capacity at the footings (kPa) and level _____

Groundwater level (pool pit 18'-3" down) _____

Sulphate exposure class _____

Lateral earth pressure of the backfill (kPa/m) _____

LOADS (NBC PART 4 / 9 - CONFIRM)

- Floors 1.9 kPa (40 psf) live; garage slab 2.4 kPa.
- Pool: 4'-6" of water (about 13.4 kPa) plus the shell; spa 3'-6" of water, raised 18".
- Roof: snow from Ss and Sr, with drifting at the lower porch roofs.
- Partitions, finishes, mechanical units, the pool dehumidifier.
- Radon protection per NBC 9.13.4 regardless of the site.

2. ELEMENTS FOR ENGINEERED DESIGN

- E1 ICF basement walls, 13'-0" tall, 11'-9" backfill: reinforcement, lintels, the exit door and window bucks; check against the ICF maker's engineered tables (S1.0, A5.0).
- E2 Wall-top support: the main floor diaphragm and the sill connection take the soil load at the top of the ICF walls - including at the stair opening and the long pool-room wall.
- E3 Footings: perimeter 24" x 10", interior bearing lines 20", steps; bearing per the geotech.
- E4 Pool and spa shells; the deepened footings round the pool room or the shell retaining them; buoyancy when empty (A5.1).
- E5 Theater depressed slab and its thickened edges (A5.1).
- E6 Exit stair areaway and light wells: 8" retaining walls, 11'-9" and 8'-0" of soil (A5.1).
- E7 Main floor framing: truss layout and spans (S1.1), the new bearing wall at x = 42.65 and the crawlspace beam at x = 103.04, the stair opening, posts and their footings.
- E8 Roof: hip trusses by the supplier; girder reactions traced down; uplift and lateral design.
- E9 Lateral: braced wall panels / shear walls for wind (and seismic) on the main floor (NBC 9.23.13).
- E10 Garage: door headers 16'-0" and 9'-0", the slab, frost walls.
- E11 Porches and deck: piers, beams, guards and their anchorage.

3. QUESTIONS FOR THE ENGINEER

- Q1 Is 8" ICF core enough for 13'-0" walls with 11'-9" of backfill, or 10"? Bar size and spacing?
- Q2 Deepen the pool-room footings, or design the pool shell to retain them?
- Q3 Does the groundwater level allow a pit 18'-3" below grade? Dewatering, drain loop, relief valve.
- Q4 Strip footings vs thickened slab at the interior bearing lines?
- Q5 Floor trusses at 18" for the 26'-11" pool-room span - spacing, vibration, moisture?
- Q6 How is the ICF wall top braced at the stair opening and the exit door?
- Q7 Can the exit-stair and light-well walls cantilever, or do they need ties to the house?
- Q8 Sulphate class and concrete mixes for the footings, walls and shells?
- Q9 Frost walls vs frost-protected shallow foundations for the crawlspace and garage?
- Q10 Field reviews: which pours need an engineer's inspection (letters of assurance)?
- Q11 Building area over 600 m2 (cover sheet): which path does the engineer recommend?

4. MATERIALS ASSUMED

Concrete (CSA A23.1, verify)

- Footings + ICF walls: S-class per geotech, else N
- Basement slab: N, 25-32 MPa, w/cm <= 0.55, wet cure 7 days
- Garage slab: C-2, 32 MPa, w/cm 0.45, air-entrained
- Pool + spa: 35 MPa, w/cm 0.40, crystalline admixture
- Frost walls, piers: as footings; air-entrained if exposed

ICF

- 8" core, 2 5/8" EPS faces (13 1/4"); brand TBD

Reinforcement

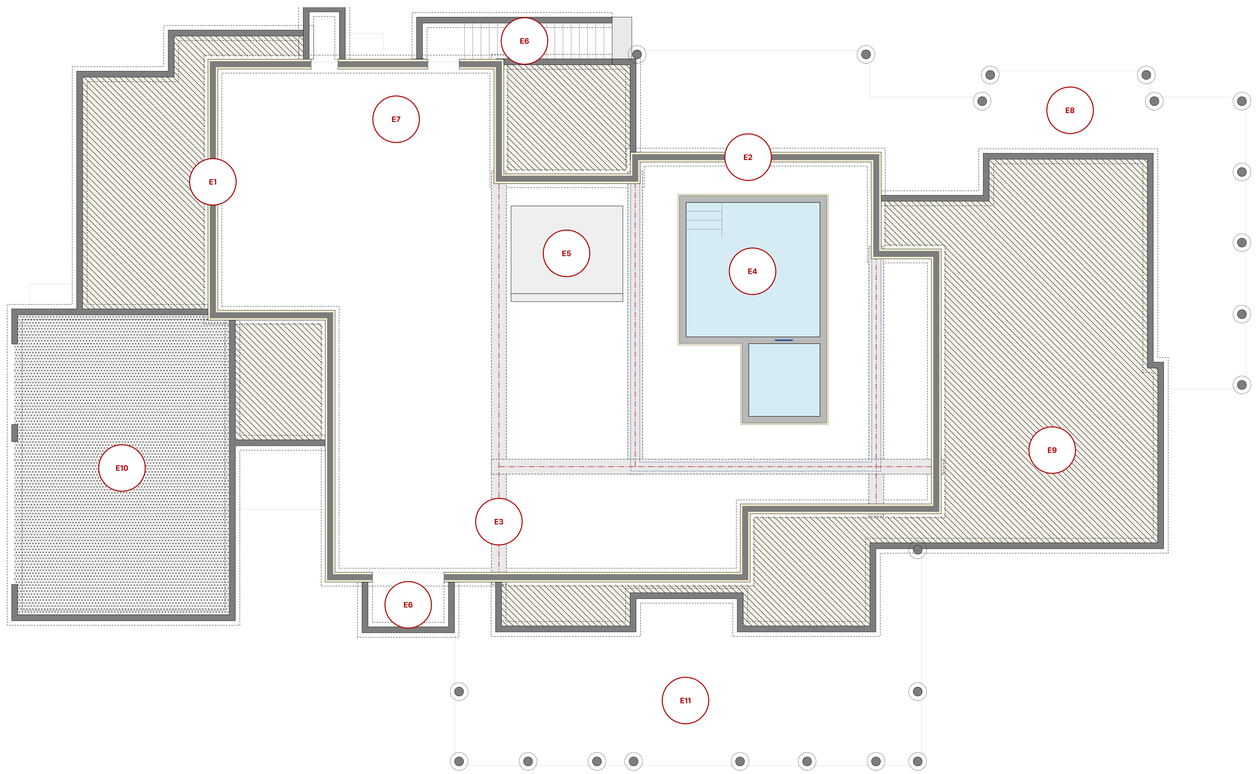
- CSA G30.18 400W; sizes by the engineer

Wood

- Floor trusses 18" (approx.), roof trusses: by supplier
- Sill plates pressure-treated on a gasket

Phase 1 embeds and sleeves

- Plumbing under the slab, sump, radon pipe, pool and spa
- piping and drains, conduits, the tunnel knock-out in
- the UTILITY room wall (future), window and door bucks



1 / S0.1 **KEY PLAN - WHERE EACH ELEMENT IS (FOUNDATION, S1.0)**

SCALE: 1/8" = 1'-0"

HOW TO USE THIS SET

This set describes the design the owners want; it is not engineered. The engineer of record (APEGA) designs the structure and returns stamped drawings or a design letter for the items above, with the truss supplier's layouts.
The permit set is then completed with the engineer's drawings, the truss layouts, the energy (9.36) compliance report, the surveyed plot plan and the permits listed in docs/permit/README.md.
Mark up the PDF or the DXF files freely; the model (models/*.scad) is the source and every sheet is regenerated from it with tools/make_drawings.py.

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SHEET TITLE

STRUCTURAL DESIGN BRIEF

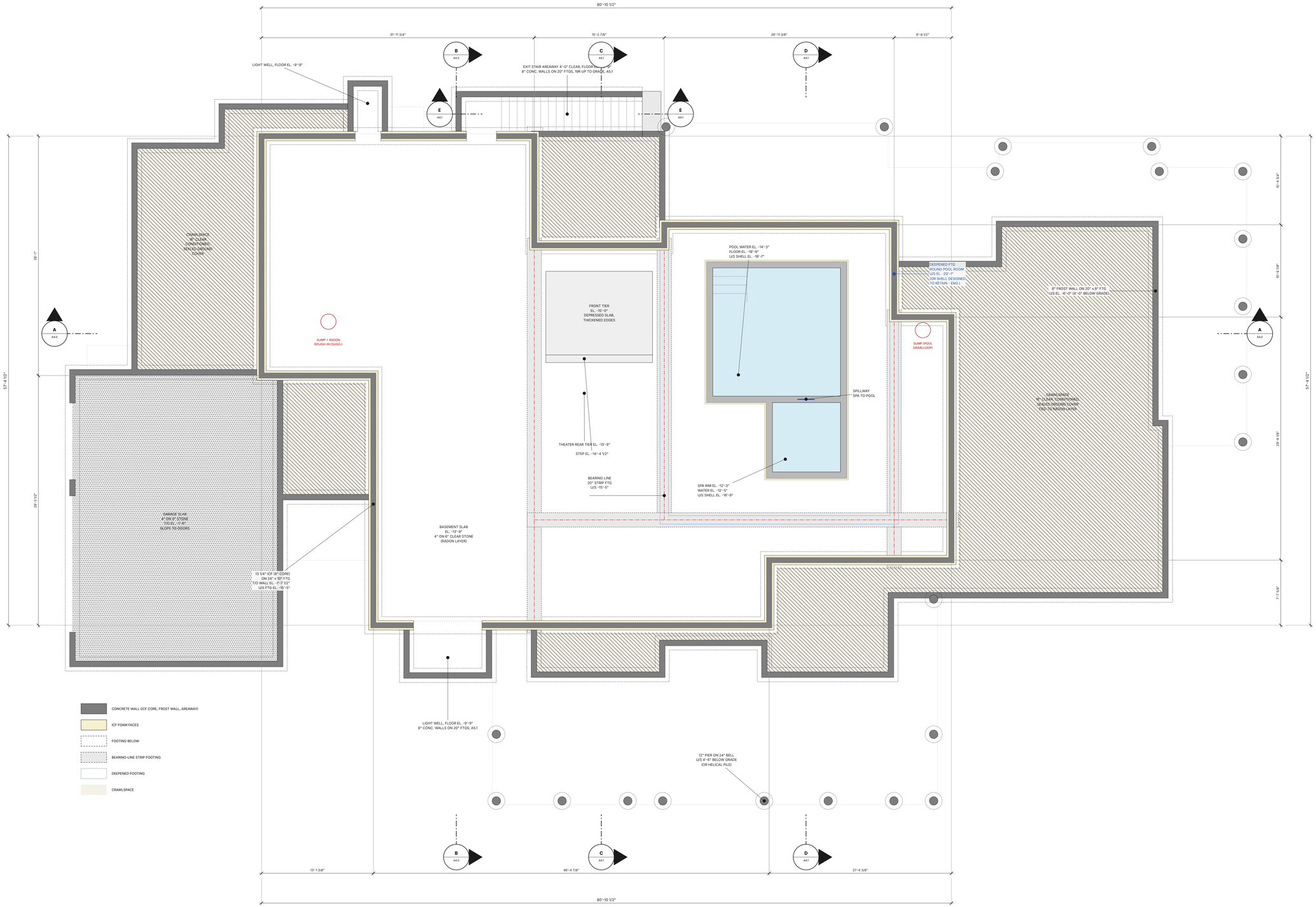
SCALE: NONE

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S0.1

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1 / S1.0

FOUNDATION PLAN

SCALE: 3/16" = 1'-0"

N

0' 4' 8' 16' 32'

FOUNDATION NOTES

- Dimensions to the ICF wall centre line (= the main floor wall centre lines). The form's outside face is 6 5/8" outside it.
- Footings bear on undisturbed soil at EL. -15'-5", about 13'-5" below grade: bearing capacity, sulphate class and groundwater from the geotechnical report.
- Reinforcement, dowels, lintels over openings and the wall-top connection: by the engineer (S0.1).

SLAB AND DRAINAGE

- 4" slab on 10 mil sealed poly on 6" clear stone (gas-permeable layer, NBC 9.13.4); 2" rigid insulation under it recommended.
- Weeping tile round the footings to the sump; the pool pit gets its own drain loop (A5.1).
- Dampproofing / waterproofing and drainage board on the ICF, free-draining backfill.

CONCRETE (CSA A23.1, verify)

- Footings, walls: S-class per the geotech, else N. Slab: N, 25-32 MPa.
- Garage slab: C-2, 32 MPa, air-entrained. Pool/spa shell: 35 MPa, w/cm 0.40 (A5.1).

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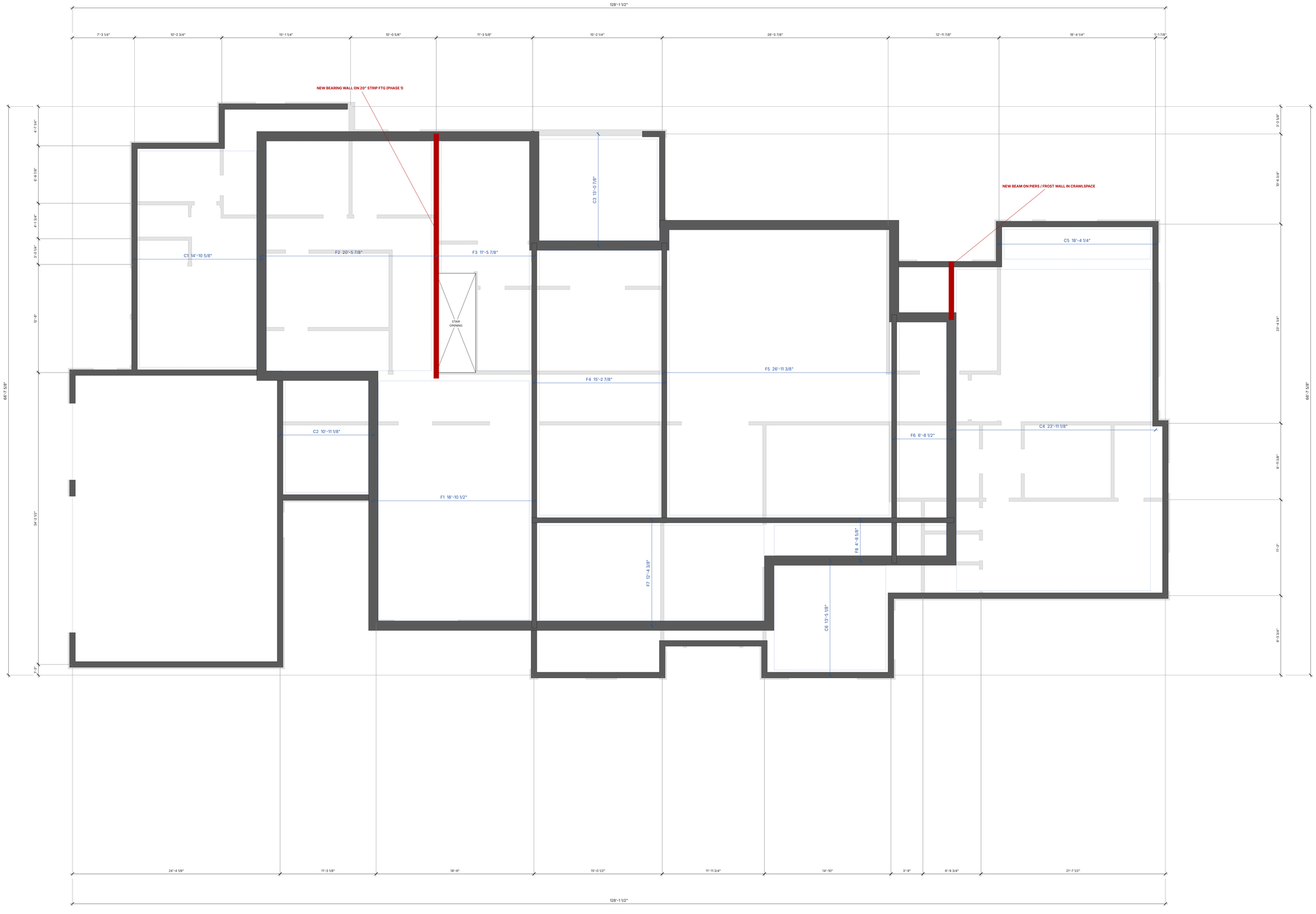
SCALE: 3/16" = 1'-0"

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S1.0

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1 / S1.1 **MAIN FLOOR FRAMING CONCEPT**

SCALE: 3/16" = 1'-0"

N

0' 4' 8' 16' 32'

FRAMING CONCEPT - FOR THE ENGINEER

- Main floor: floor trusses spanning between the supports shown dark: ICF walls, interior bearing lines on strip footings, frost walls.
- Blue: indicative clear spans (F = over the basement, C = over crawlspace). Depth allowed: 1'-7 1/2" from T/O ICF to T/O subfloor.
- Two supports are added (red): a bearing wall on x = 42.65 so the west zone is not a 32 ft span, and a crawlspace beam line on x = 103.04 under the office.

CHECK

- F5 pool room: 26'-11" clear over a humid room - deep or closely spaced trusses, and protect them from the pool air (vapour control in the ceiling).
- Stair opening 4'-7 1/2" x 11'-7 5/8": headers and trimmers. Garage door headers 16'-0" and 9'-0".
- Roof: hip trusses by the truss designer. Girder trusses where the hips meet put point loads on interior walls - trace them to the footings.

BEARING WALLS ARE PHASE 1

- Everything that carries the main floor (bearing walls, beams, posts and their footings) is built with the house, even where the basement stays unfinished until Phase 2.

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ENGINEER'S REVIEW / SEAL

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MAIN FLOOR FRAMING CONCEPT

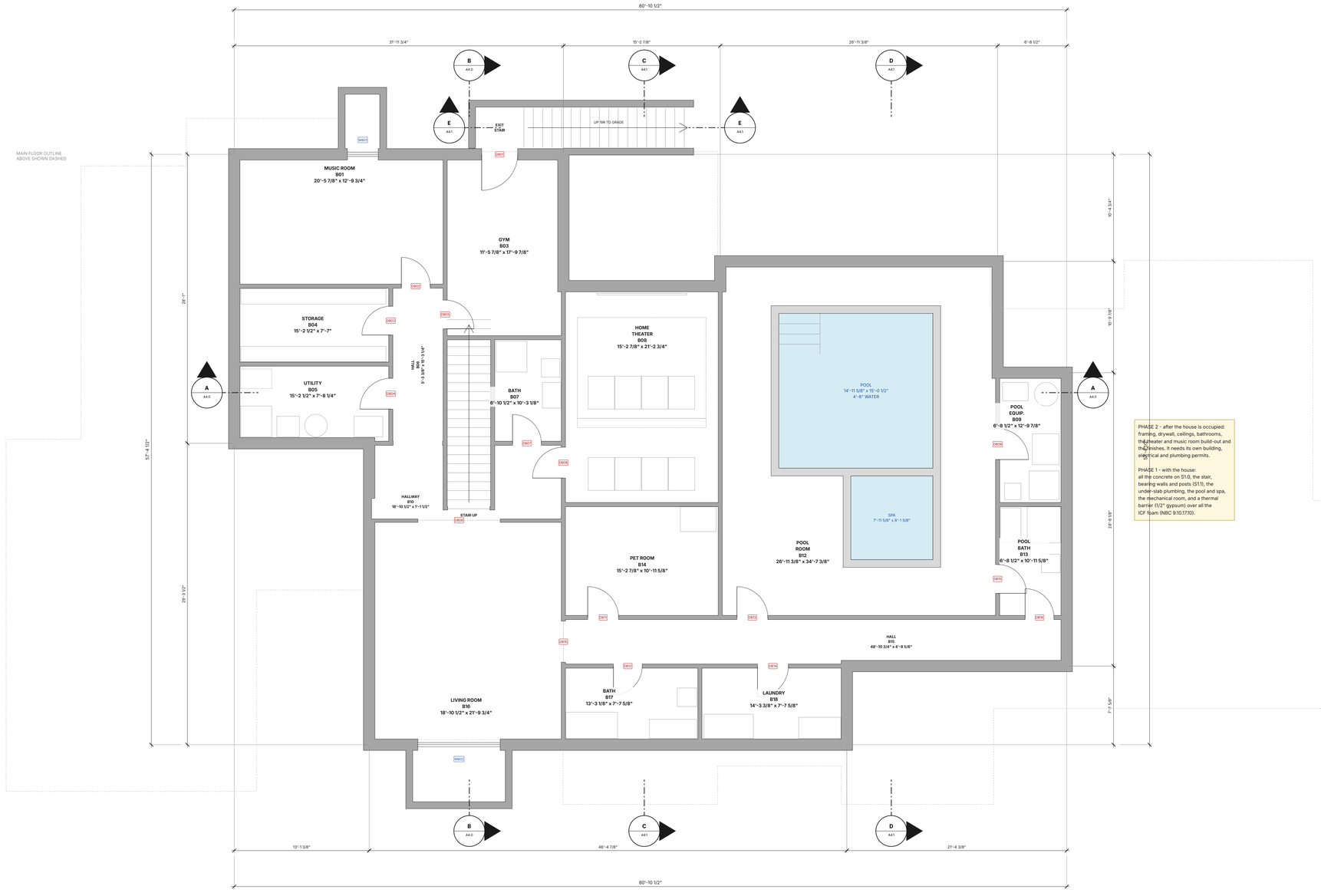
SCALE: 3/16" = 1'-0"

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S1.1

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PHASE 2 - after the house is occupied: framing, drywall, ceilings, bathrooms, the theater and music room build-out and the finishes. It needs its own building, electrical and plumbing permits.

PHASE 1 - with the house: all the concrete on S1.0, the stair, bearing walls and posts (S1.1), the under-slab plumbing, the pool and spa, the mechanical room, and a thermal barrier (1/2" gypsum) over all the ICF foam (NBC 9.10.17.10).

1 BASEMENT PLAN

A1.1

SCALE: 3/16" = 1'-0"

N

0' 4' 8' 16' 32'

PLAN NOTES

- Basement floor EL. -13'-9", 12'-0" ceilings to the underside of the main floor framing.
- Theater: rear tier at floor level, one 7 1/2" step, front tier 1'-3" down (A5.1).
- Pool 15'-0" x 15'-0", 4'-6" water; spa raised 18" (A5.1).

EXITS AND LIGHT

- No bedrooms in the basement, so no egress windows are needed (NBC 9.9.10).
- Exits: the main stair and the exterior exit stair from the gym (A5.1).
- Windows WB01-WB02 sit in light wells with 42" guards (A5.1).

BEFORE THE SLAB (PHASE 1)

- Under-slab drains for every future basement fixture, floor drains, the sump, the radon pipe, the pool and spa piping and the hydrostatic relief valve.

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BASEMENT PLAN

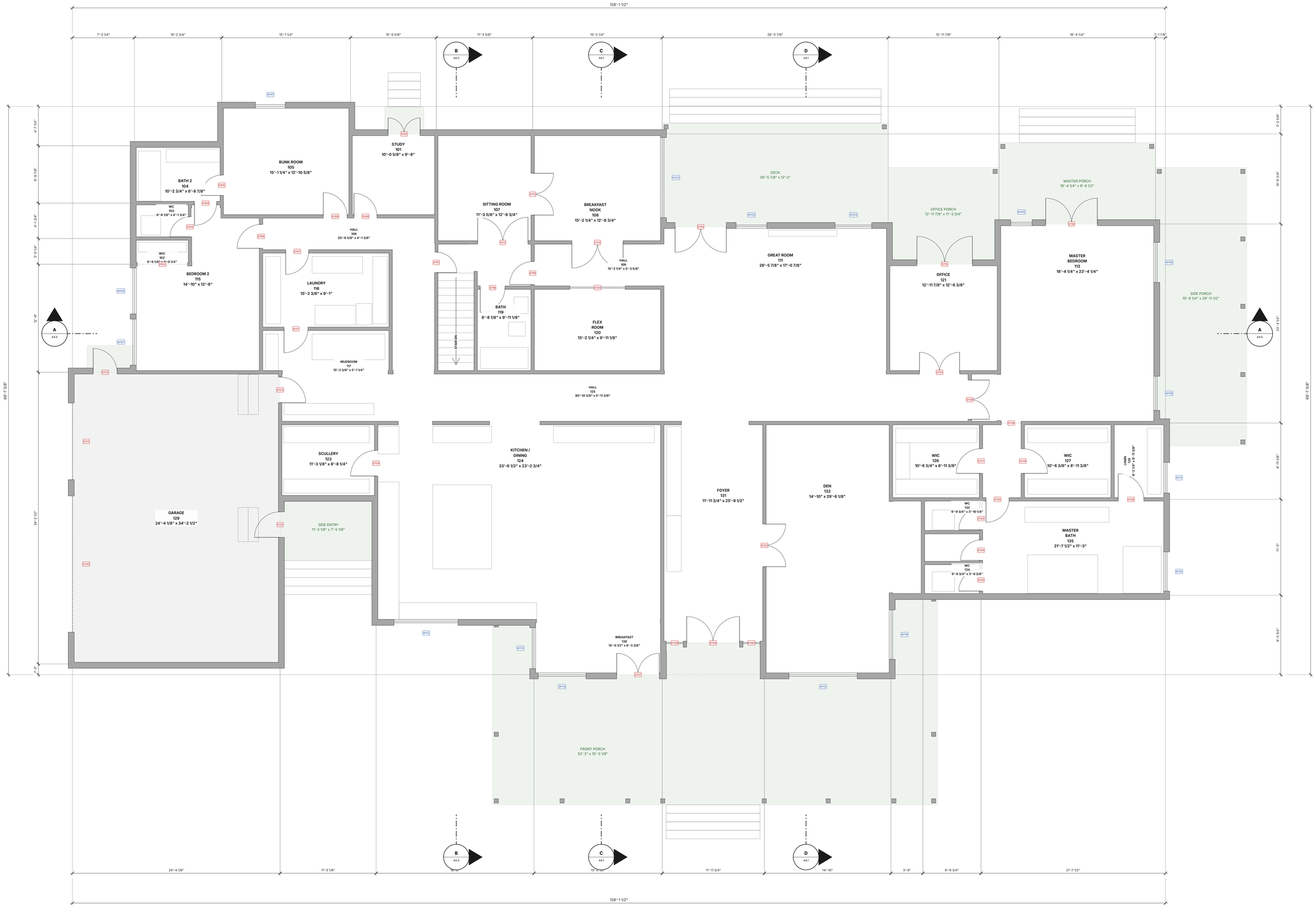
SCALE: 3/16" = 1'-0"

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A1.1

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1 / A1.2 MAIN FLOOR PLAN

SCALE: 3/16" = 1'-0"

N

0' 4' 8' 16' 32'

PLAN NOTES

- Room tags: name, number (A6.0), size between wall centre lines.
- Door marks D1xx, window marks W1xx: see the schedules on A6.0.
- Porch floors are 2'-0" (610 mm) above grade; guards (NBC 9.8.8) unless the finished grade is raised to within 600 mm of the porch floor.

LIFE SAFETY

- Smoke alarms in every bedroom, the hall outside them and on each storey: hard-wired, interconnected (NBC 9.10.19).
- CO alarms near the sleeping areas: the garage is attached (NBC 9.32.3.9).
- Garage door to the house: self-closing, weatherstripped; garage walls and ceiling gas-proofed (NBC 9.10.13.15, 9.10.9.16).

STAIRS AND STEPS

- Main stair: 22 risers x 7 1/2", treads 10 1/2", headroom 1 993 mm min. (A5.1).
- Porch steps (4 risers of 6") and garage steps (3 risers): handrails per NBC 9.8.7; check the garage door landing (9.8.6).

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MAIN FLOOR PLAN

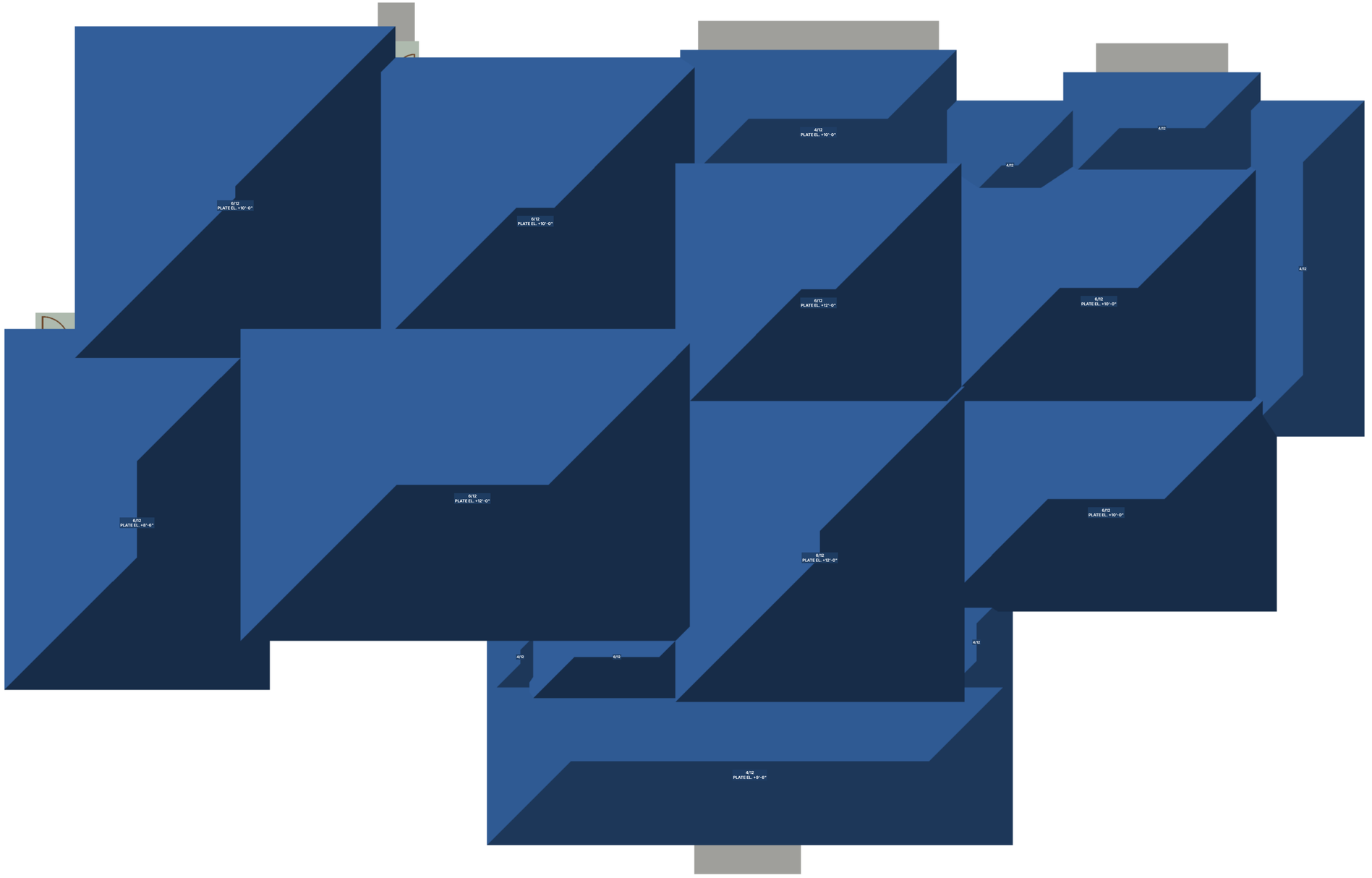
SCALE: 3/16" = 1'-0"

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A1.2

DATE 2026-11-03

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ROOF PLAN

1 / A1.3

SCALE: 3/16" = 1'-0"

N

0' 4' 8' 16' 32'

ROOF NOTES

- Hip roofs: 6/12 over the house, 4/12 over the porches; 1'-6" overhangs on the house, 1'-0" on the porches. Plate heights as tagged (main floor = 0'-0").
- Engineered roof trusses by the truss supplier, designed for the site's snow and wind loads (S0.1). Raised-heel (energy-heel) trusses for full attic insulation at the walls.

FOR THE ENGINEER

- Where the hip blocks meet, valleys and girder trusses bring point loads onto interior walls and headers: take them down through the main floor framing to the footings (S1.1).
- Snow drifting at the lower porch roofs against the higher house roofs (NBC 4.1.6 / 9.4.2).

ENVELOPE (NBC 9.36, verify zone 7A/7B)

- Attic insulation, ventilation (1/300) and an air barrier per 9.36 and 9.19; the energy advisor's model sets the final values (performance path).

NEW SINGLE-FAMILY DWELLING

Attached garage, full ICF basement, indoor pool + spa

Rural Alberta - legal land description: TBD

Owner / applicant: _____

NOT FOR CONSTRUCTION

FOR ENGINEERING REVIEW AND PERMIT PREPARATION

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Elevations are relative to the main floor = 0'-0". Finished grade is taken at -2'-0" all round until a site survey and grading plan fix it.

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ENGINEER'S REVIEW / SEAL

SHEET TITLE

ROOF PLAN

SCALE: 3/16" = 1'-0"

SHEET

8 of 14

A1.3

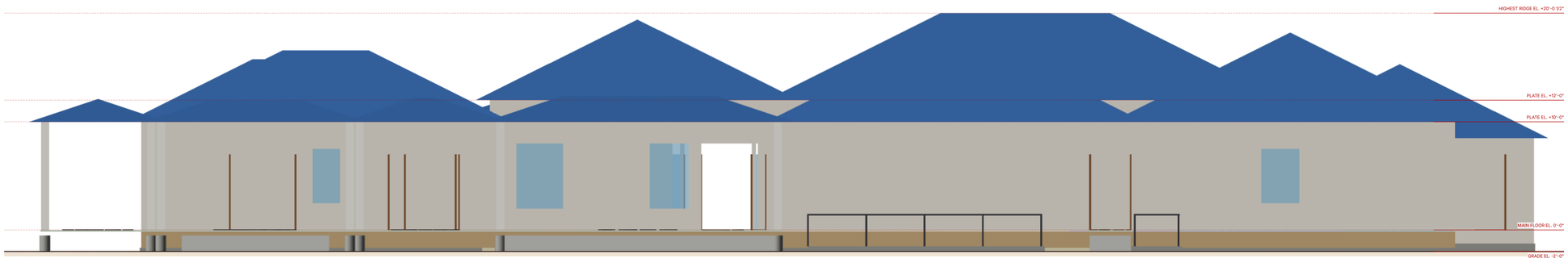
DATE 2026-11-03

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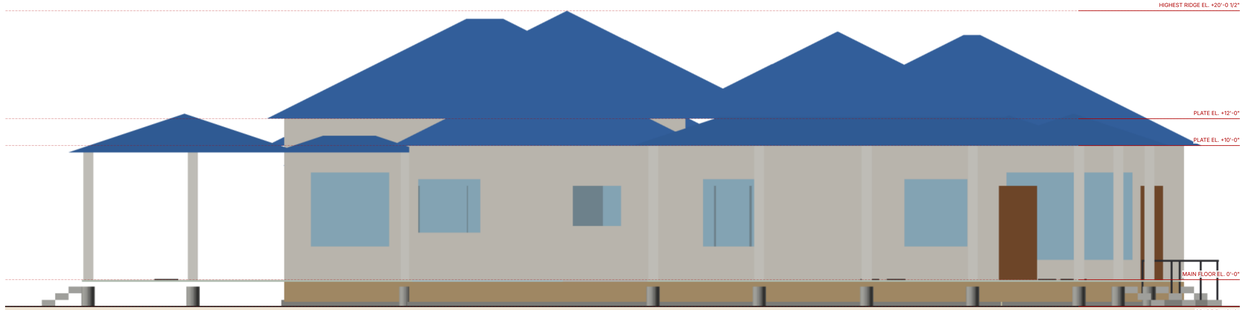
1 A3.0 **SOUTH (FRONT) ELEVATION**

SCALE: 3/16" = 1'-0"



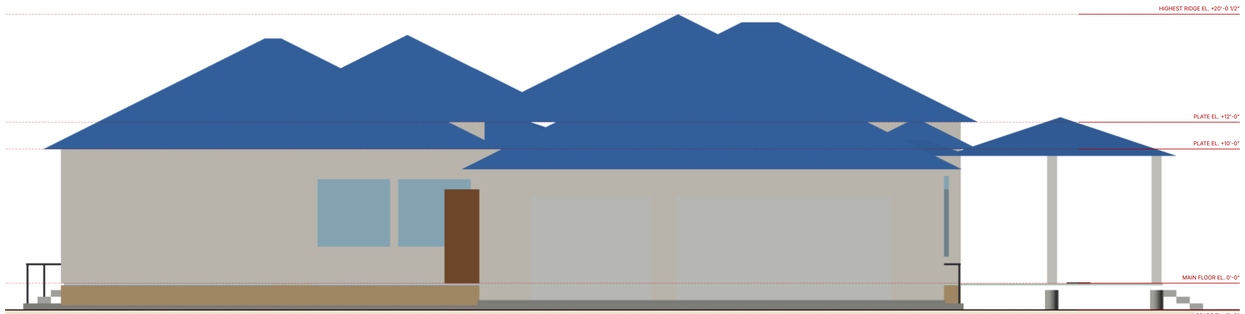
2 A3.0 **NORTH (REAR) ELEVATION**

SCALE: 3/16" = 1'-0"



3 A3.0 **EAST ELEVATION**

SCALE: 3/16" = 1'-0"



4 A3.0 **WEST ELEVATION**

SCALE: 3/16" = 1'-0"

ELEVATION NOTES

- Views are orthographic renders of the model, at scale. Finished grade is drawn at -2'-0"; the grading plan sets the real grades (slope away from the house).
- Cladding, roofing and trim materials are not selected yet; the envelope must meet NBC 9.36 (energy) and 9.27 (cladding) and suit the site's hail exposure.
- Window and door sizes: A6.0. Glazing area per wall face counts toward the spatial separation limits (NBC 9.10.15) if the house is near a property line - not expected on a rural lot.
- Highest ridge about 20'-0 1/2" above the main floor (22'-0 1/2" above grade): check the land-use bylaw height limit.
- Porch floors 2'-0" above grade: guards or raised grade (A1.2).
- The model draws door leaves open, so some doorways show as openings.

NEW SINGLE-FAMILY DWELLING

Attached garage, full ICF basement, indoor pool + spa

Rural Alberta - legal land description: TBD

Owner / applicant: _____

NOT FOR CONSTRUCTION

FOR ENGINEERING REVIEW AND PERMIT PREPARATION

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ENGINEER'S REVIEW / SEAL

SHEET TITLE

ELEVATIONS

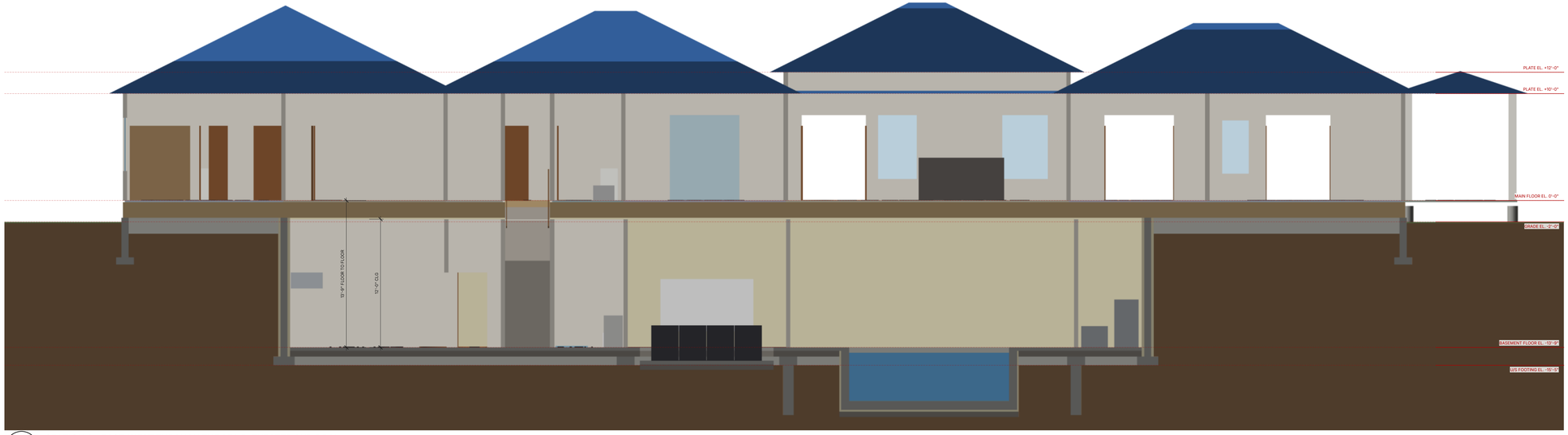
SCALE: 3/16" = 1'-0"

SHEET 9 of 14

A3.0

DATE 2026-11-03

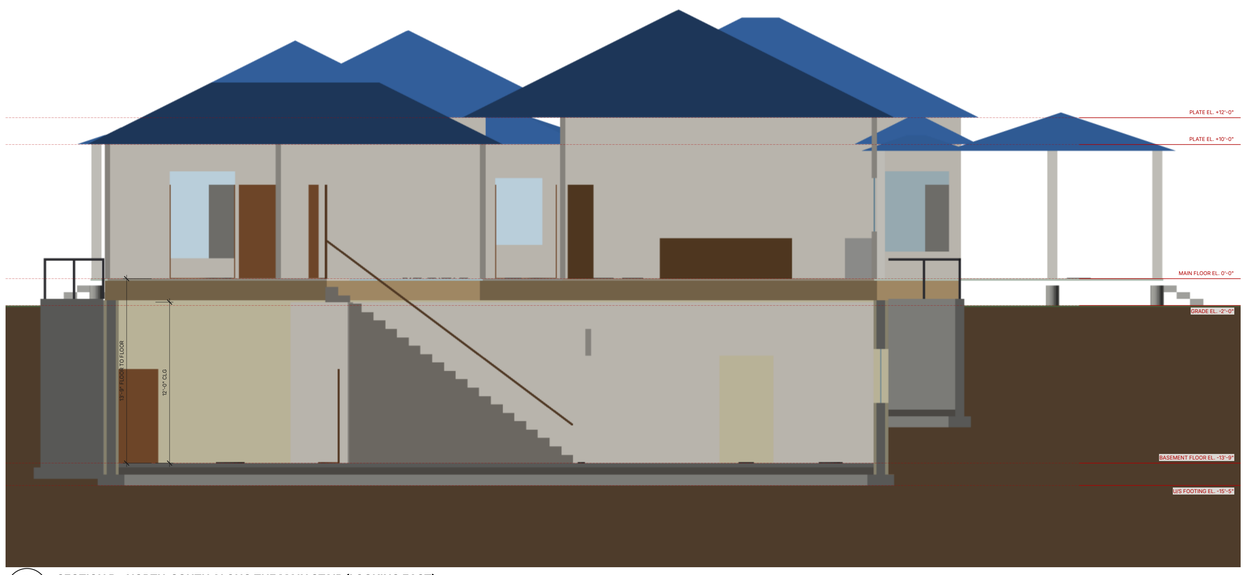
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A
A4.0

SECTION A - EAST-WEST THROUGH THE STAIR, THEATER AND POOL

SCALE: 3/16" = 1'-0"



B
A4.0

SECTION B - NORTH-SOUTH ALONG THE MAIN STAIR (LOOKING EAST)

SCALE: 3/16" = 1'-0"

SECTION NOTES

- Sections are cut through the model and drawn at scale; cut faces are darker than what is seen beyond. Earth is drawn round the foundation; grade at -2'-0".
- Main floor 0'-0"; basement floor EL. -13'-9"; floor to floor 13'-9"; 12'-0" basement ceilings.
- ICF walls stand about 12'-11 1/2" from the footing to the sill, with 11'-9" of backfill above the basement floor: beyond prescriptive ICF tables, engineered design required (S0.1).
- The ICF wall tops are held by the main floor: its diaphragm and the sill connection must take the soil load at the top of the wall (engineer).
- Plate heights 10'-0" and 12'-0" (8'-6" garage); hip roofs at 6/12.
- Section A passes the stair opening: the headroom check is on A5.1.

NEW SINGLE-FAMILY DWELLING

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ENGINEER'S REVIEW / SEAL

SHEET TITLE

BUILDING SECTIONS A AND B

SCALE: 3/16" = 1'-0"

SHEET 10 of 14

A4.0

DATE 2026-11-03

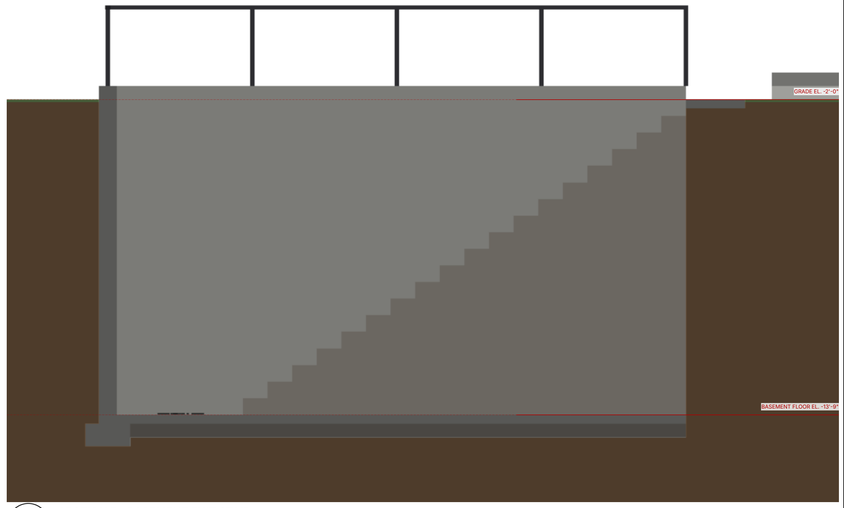
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C
A4.1

SECTION C - THROUGH THE THEATER (LOOKING EAST)

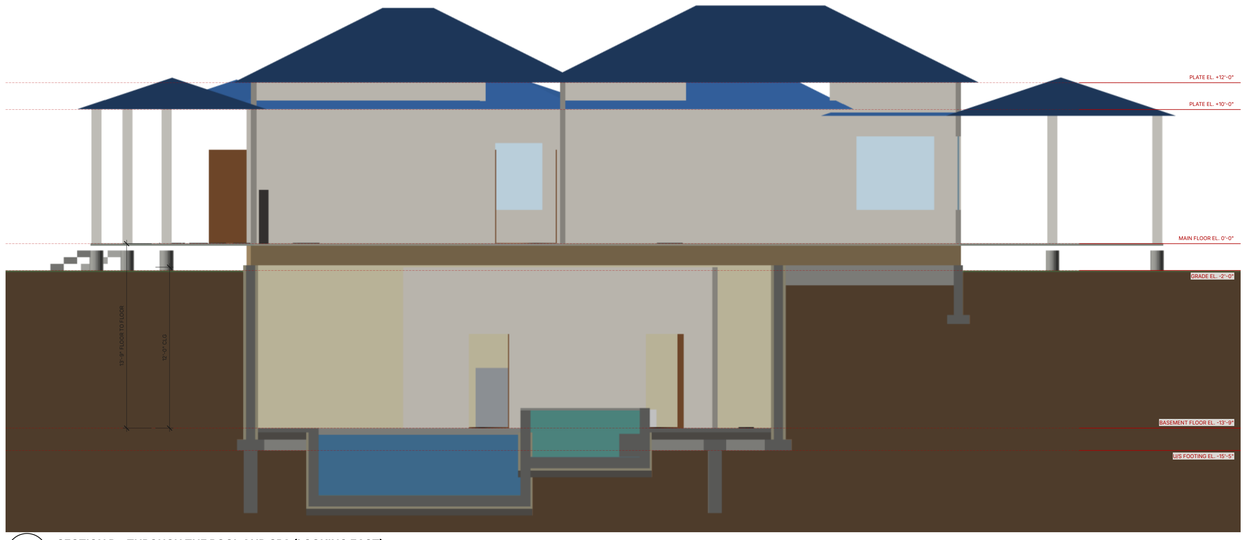
SCALE: 3/16" = 1'-0"



E
A4.1

SECTION E - BASEMENT EXIT STAIR

SCALE: 3/8" = 1'-0"



D
A4.1

SECTION D - THROUGH THE POOL AND SPA (LOOKING EAST)

SCALE: 3/16" = 1'-0"

NOTES

- C: the stepped theater floor - rear tier at the basement floor, one 7 1/2" step, front tier 15" down; the lowered area stops 1'-0" short of the side walls and 2'-6" short of the screen wall, outside the footings' 1:1 bearing zones (A5.1).
- D: pool and spa shells, the deepened footings round the pool room and the pit's insulation and stone (A5.1).
- E: the exit stair, 19 risers up to grade along the gym's north wall, in an areaway with 8" concrete walls and a 42" guard (A5.1).

NEW SINGLE-FAMILY DWELLING

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Owner / applicant: _____

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ENGINEER'S REVIEW / SEAL

SHEET TITLE

BUILDING SECTIONS C, D AND E

SCALE: 3/16" = 1'-0" U.N.O.

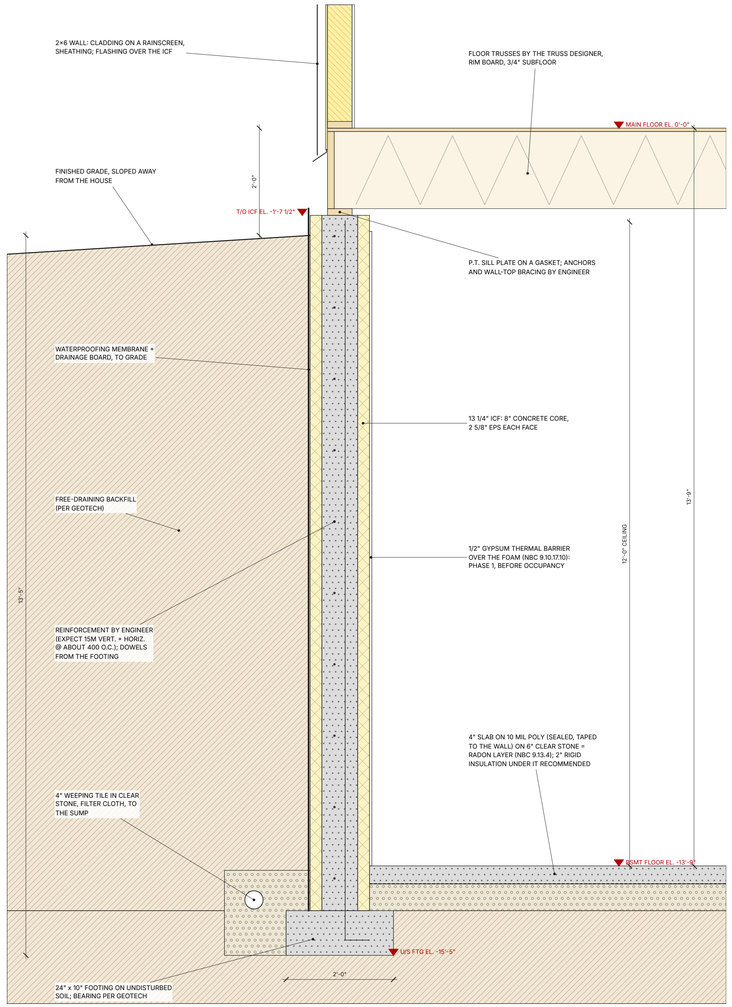
SHEET

11 of 14

A4.1

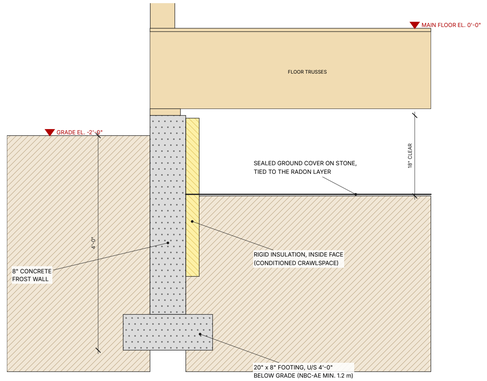
DATE 2026-11-03

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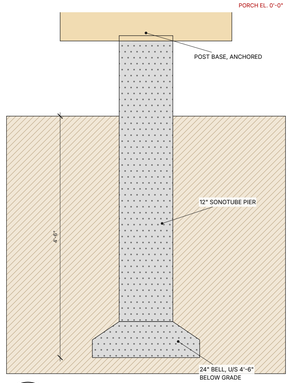
1 A5.0 **TYPICAL ICF BASEMENT WALL**

SCALE: 3/4" = 1'-0"



2 A5.0 **FROST WALL AT CRAWLSPACE**

SCALE: 3/4" = 1'-0"



4 A5.0 **PORCH PIER**

SCALE: 3/4" = 1'-0"

5 A5.0 **GARAGE SLAB EDGE**

SCALE: 3/4" = 1'-0"

3 A5.0 **INTERIOR BEARING LINE FOOTING**

SCALE: 3/4" = 1'-0"

FOUNDATION - FOR THE ENGINEER

Design items (S0.1)

- ICF walls: about 13'-0" tall with 11'-9" of backfill - beyond prescriptive tables. Vertical and horizontal steel, wall-top support by the floor, lintels over the door and windows.
- Footings: size from the geotech's bearing capacity; steps where the pool room footings deepen.
- Interior bearing lines: strip footings for the line loads from the main floor and the roof's girder trusses (S1.1).
- Areaway and light wells: 8" cantilevered retaining walls, 11'-9" and 8'-0" of soil.

Assumed until the engineer and geotech say otherwise

- Concrete: footings and walls S-class (sulphate) per the geotech, else N.
- Frost: footings 4'-0" below grade where there is no basement (NBC-AE minimum 1.2 m).
- Radon: sealed 10 mil poly, 6" clear stone, a capped pipe through the slab, NBC 9.13.4.
- Thermal barrier: 1/2" gypsum over all interior foam before occupancy, even with the basement unfinished (NBC 9.10.17.10).

Inspections

- Footing and wall inspections by the authority having jurisdiction, and field reviews by the engineer of the reinforcement before each pour.

NEW SINGLE-FAMILY DWELLING

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ENGINEER'S REVIEW / SEAL

SHEET TITLE

DETAILS - FOUNDATION

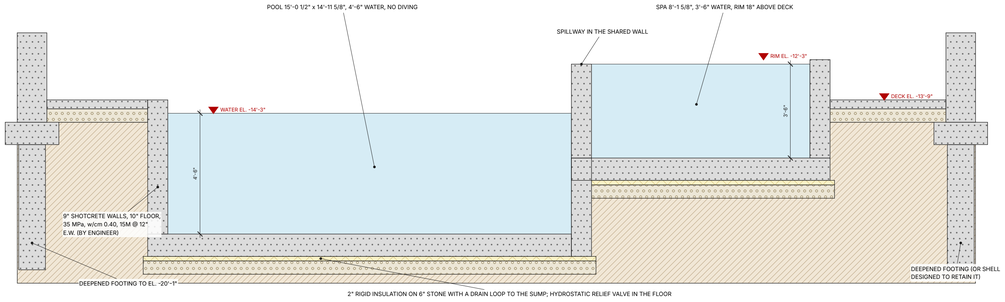
SCALE: AS NOTED

SHEET 12 of 14

A5.0

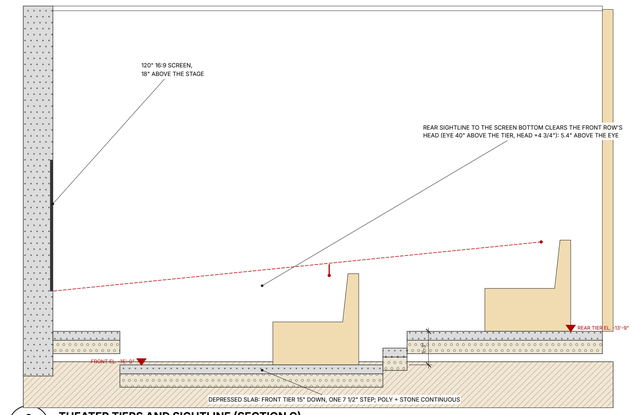
DATE 2026-11-03

REV. 0 - issued for review



1 A5.1 **POOL AND SPA (SECTION D)**

SCALE: 3/8" = 1'-0"



2 A5.1 **THEATER TIERS AND SIGHTLINE (SECTION C)**

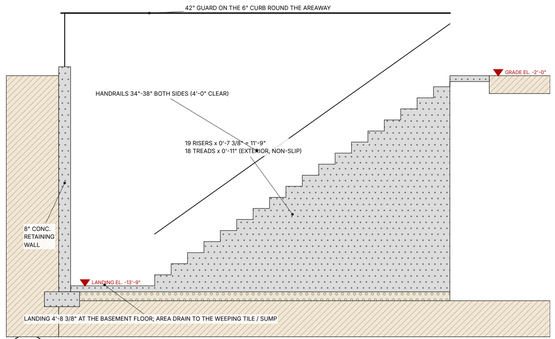
SCALE: 3/8" = 1'-0"

POOL AND SPA - FOR THE ENGINEER

- Shell: wet-mix shotcrete (ACI 506.2), 35 MPa at 28 days, w/cm 0.40, crystalline waterproofing admixture; one continuous placement for the pool, the spa and the shared wall.
- Design the shell for the water inside and the soil and groundwater outside, empty and full; uplift when empty (hydrostatic relief valve, drain loop).
- Footings round the pool room are deepened below the pit, or the shell retains them.
- Two main drains per pool and spa (anti-entrapment), bonding grid, GFCI electrical permit (CEC Section 68).
- A dedicated pool dehumidifier; the room's envelope and the trusses above protected from the pool air.

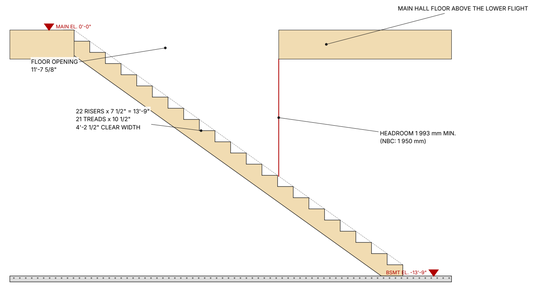
THEATER

- Depressed slab poured with the basement floor (Phase 1); the seating, room-within-a-room walls and the screen are Phase 2.



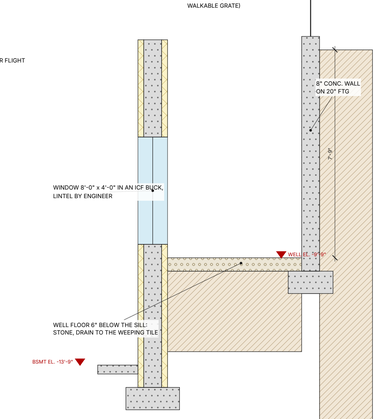
3 A5.1 **BASEMENT EXIT STAIR (SECTION E)**

SCALE: 1/4" = 1'-0"



5 A5.1 **MAIN STAIR (SECTION B)**

SCALE: 1/4" = 1'-0"



4 A5.1 **LIGHT WELL (LIVING ROOM)**

SCALE: 3/8" = 1'-0"

NEW SINGLE-FAMILY DWELLING

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Owner / applicant: _____

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FOR ENGINEERING REVIEW AND PERMIT PREPARATION

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ENGINEER'S REVIEW / SEAL

SHEET TITLE

DETAILS - POOL, THEATER, STAIRS, LIGHT WELL

SCALE: AS NOTED

SHEET 13 of 14

A5.1

DATE 2026-11-03

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DOOR SCHEDULE (1 of 2)

MARK	LVL	BETWEEN	TYPE	WIDTH	HEIGHT	SW.	NOTES
D101	MAIN	STUDY / PORCH	DOUBLE SWING	3'-9"	8'-0"	N	EXTERIOR: INSULATED, WEATHERSTRIPPED
D102	MAIN	BEDROOM 2 / WIC	CASED OPENING	5'-10 3/4"	8'-0"		
D103	MAIN	WC / HALL	SWING	2'-4 1/4"	7'-0"	W	
D104	MAIN	HALL / BATH 2	SWING	2'-7 1/4"	7'-0"	S	
D105	MAIN	BATH 2 / BUNK ROOM	SWING	2'-4 3/4"	7'-0"	W	
D106	MAIN	HALL / HALL	SWING	2'-9 5/8"	7'-0"	W	
D107	MAIN	LAUNDRY / HALL	SWING	2'-8 3/8"	7'-0"	S	
D108	MAIN	HALL / BUNK ROOM	SWING	2'-9 5/8"	7'-0"	N	
D109	MAIN	HALL / STUDY	SWING	2'-8 3/8"	7'-0"	N	
D110	MAIN	HALL / HALL	SWING	2'-4 3/4"	7'-0"	E	
D111	MAIN	HALL / SITTING ROOM	DOUBLE SWING	5'-10 3/4"	8'-0"	N	
D112	MAIN	SITTING ROOM / BREAKFAST NOOK	DOUBLE SWING	4'-10 3/4"	8'-0"	E	
D113	MAIN	HALL / BREAKFAST NOOK	DOUBLE SWING	6'-0"	8'-0"	S	
D114	MAIN	GREAT ROOM / DECK	DOUBLE SWING	6'-0"	8'-0"	S	EXTERIOR: INSULATED, WEATHERSTRIPPED
D115	MAIN	OFFICE / OFFICE PORCH	DOUBLE SWING	6'-6"	8'-0"	N	EXTERIOR: INSULATED, WEATHERSTRIPPED
D116	MAIN	MASTER BEDROOM / MASTER PORCH	DOUBLE SWING	6'-0 5/8"	8'-0"	N	EXTERIOR: INSULATED, WEATHERSTRIPPED
D117	MAIN	MUDROOM / LAUNDRY	SWING	2'-9 5/8"	7'-0"	S	
D118	MAIN	BATH / HALL	SWING	2'-10 3/4"	7'-0"	S	
D119	MAIN	HALL / HALL	SWING	2'-8 3/8"	7'-0"	W	
D120	MAIN	FLEX ROOM / HALL	SLIDING (GLAZED)	6'-6"	8'-0"		
D121	MAIN	OUTSIDE / GARAGE	OVERHEAD	9'-0"	8'-0"		HEADER BY ENGINEER
D122	MAIN	GARAGE / PORCH	SWING	2'-9 5/8"	7'-0"	N	EXTERIOR: INSULATED, WEATHERSTRIPPED
D123	MAIN	GARAGE / HALL	SWING	3'-0"	7'-0"	E	TO GARAGE: SELF-CLOSING, WEATHERSTRIPPED (9.10.13.15)
D124	MAIN	SCULLERY / KITCHEN / DINING	SWING	3'-0"	7'-0"	W	
D125	MAIN	HALL / OFFICE	DOUBLE SWING	4'-8 3/8"	8'-0"	N	
D126	MAIN	HALL / HALL	DOUBLE SWING	4'-7 1/4"	8'-0"	E	
D127	MAIN	WIC / HALL	SWING	2'-10 3/4"	7'-0"	W	
D128	MAIN	HALL / MASTER BEDROOM	CASED OPENING	2'-3 5/8"	8'-0"		
D129	MAIN	HALL / WIC	SWING	2'-10 3/4"	7'-0"	E	

DOOR SCHEDULE (2 of 2)

MARK	LVL	BETWEEN	TYPE	WIDTH	HEIGHT	SW.	NOTES
D130	MAIN	OUTSIDE / GARAGE	OVERHEAD	16'-0"	8'-0"		HEADER BY ENGINEER
D131	MAIN	GARAGE / SIDE ENTRY	SWING	3'-0"	7'-0"	W	EXTERIOR: INSULATED, WEATHERSTRIPPED
D132	MAIN	FOYER / DEN	DOUBLE SWING	5'-0"	8'-0"	E	
D133	MAIN	WC / MASTER BATH	SWING	2'-7 1/4"	7'-0"	W	
D134	MAIN	CLOSET / MASTER BATH	SWING	2'-4 3/4"	7'-0"	W	
D135	MAIN	MASTER BATH / HALL	SWING	2'-8 3/8"	7'-0"	S	
D136	MAIN	MASTER BATH / LINEN	SWING	3'-0"	7'-0"	N	
D137	MAIN	FRONT PORCH / BREAKFAST	DOUBLE SWING	5'-0"	8'-0"	N	EXTERIOR: INSULATED, WEATHERSTRIPPED
D138	MAIN	PORCH / FOYER	SIDELIGHT (GLAZED)	2'-0"	8'-0"		EXTERIOR: INSULATED, WEATHERSTRIPPED, SAFETY GLAZING
D139	MAIN	PORCH / FOYER	DOUBLE SWING	6'-2 3/8"	8'-0"	N	EXTERIOR: INSULATED, WEATHERSTRIPPED
D140	MAIN	PORCH / FOYER	SIDELIGHT (GLAZED)	2'-0"	8'-0"		EXTERIOR: INSULATED, WEATHERSTRIPPED, SAFETY GLAZING
D141	MAIN	WC / MASTER BATH	SWING	2'-7 1/4"	7'-0"	W	
D801	BSMT	GYM / EXIT STAIR	SWING	3'-6"	7'-0"	S	EXTERIOR: INSULATED, WEATHERSTRIPPED; TO THE EXIT STAIR, OPENS IN
D802	BSMT	HALL / MUSIC ROOM	SWING	2'-9 5/8"	7'-0"	N	
D803	BSMT	STORAGE / HALL	SWING	2'-9 5/8"	7'-0"	W	
D804	BSMT	UTILITY / HALL	SWING	3'-0"	7'-0"	W	
D805	BSMT	HALL / GYM	SWING	2'-9 5/8"	7'-0"	E	
D806	BSMT	LIVING ROOM / HALLWAY	CASED OPENING	8'-0"	8'-0"		
D807	BSMT	HALL / BATH	SWING	2'-9 5/8"	7'-0"	N	
D808	BSMT	HALL / HOME THEATER	SWING	3'-0"	7'-0"	W	SOLID CORE, SEALS
D809	BSMT	POOL ROOM / POOL EQUIP.	SWING	3'-0"	7'-0"	E	
D810	BSMT	LIVING ROOM / HALL	CASED OPENING	4'-0"	8'-0"		
D811	BSMT	HALL / PET ROOM	SWING	3'-0"	7'-0"	N	
D812	BSMT	BATH / HALL	SWING	2'-9 5/8"	7'-0"	S	
D813	BSMT	HALL / POOL ROOM	SWING	3'-0"	7'-0"	N	
D814	BSMT	LAUNDRY / HALL	SWING	3'-0"	7'-0"	S	
D815	BSMT	POOL ROOM / POOL BATH	SWING	2'-9 5/8"	7'-0"	E	
D816	BSMT	HALL / POOL BATH	SWING	2'-9 5/8"	7'-0"	N	

WINDOW SCHEDULE

MARK	LVL	ROOM	WIDTH	HEIGHT	SILL	HEAD	m2	NOTES
W101	MAIN	BUNK ROOM	3'-6"	5'-0"	2'-6"	7'-6"	1.63	BEDROOM: EGRESS 0.35 m2 MIN., NO SIDE < 380 mm (9.9.10.1)
W102	MAIN	BREAKFAST NOOK	9'-4 3/4"	6'-6"	1'-6"	8'-0"	5.68	
W103	MAIN	GREAT ROOM	3'-7 1/4"	6'-0"	2'-0"	8'-0"	2.01	
W104	MAIN	GREAT ROOM	4'-3 5/8"	6'-0"	2'-0"	8'-0"	2.40	
W105	MAIN	MASTER BEDROOM	2'-6"	5'-0"	2'-6"	7'-6"	1.16	BEDROOM: EGRESS 0.35 m2 MIN., NO SIDE < 380 mm (9.9.10.1)
W106	MAIN	MASTER BEDROOM	4'-8 3/8"	5'-0"	2'-6"	7'-6"	2.18	BEDROOM: EGRESS 0.35 m2 MIN., NO SIDE < 380 mm (9.9.10.1)
W107	MAIN	BEDROOM 2	5'-4 3/4"	5'-0"	2'-9"	7'-9"	2.51	BEDROOM: EGRESS 0.35 m2 MIN., NO SIDE < 380 mm (9.9.10.1)
W108	MAIN	BEDROOM 2	5'-4 3/4"	5'-0"	2'-9"	7'-9"	2.51	BEDROOM: EGRESS 0.35 m2 MIN., NO SIDE < 380 mm (9.9.10.1)
W109	MAIN	MASTER BEDROOM	4'-0"	5'-0"	2'-6"	7'-6"	1.86	BEDROOM: EGRESS 0.35 m2 MIN., NO SIDE < 380 mm (9.9.10.1)
W110	MAIN	MASTER BATH	4'-7 1/4"	4'-0"	3'-6"	7'-6"	1.71	PRIVACY / SAFETY GLAZING NEAR TUB OR SHOWER
W111	MAIN	LINEN	3'-7 1/4"	3'-0"	4'-0"	7'-0"	1.00	
W112	MAIN	KITCHEN / DINING	7'-6"	4'-0"	3'-6"	7'-6"	2.79	
W113	MAIN	PORCH	4'-9 5/8"	6'-0"	2'-0"	8'-0"	2.68	
W114	MAIN	FRONT PORCH	5'-7 1/4"	6'-0"	2'-0"	8'-0"	3.12	
W115	MAIN	FRONT PORCH	8'-0"	5'-6"	2'-6"	8'-0"	4.09	
W116	MAIN	DEN	5'-9 5/8"	5'-6"	2'-6"	8'-0"	2.96	
WB01	BSMT	MUSIC ROOM	3'-0"	4'-0"	4'-6"	8'-6"	1.11	IN A LIGHT WELL (A5.1)
WB02	BSMT	LIVING ROOM	8'-0"	4'-0"	4'-6"	8'-6"	2.97	IN A LIGHT WELL (A5.1)

ROOM SCHEDULE (1 of 2)

NO.	LVL	ROOM	SIZE (C/L)	ft2	m2	CEILING	PHASE
101	MAIN	STUDY	10'-0 5/8" x 9'-8"	97	9.0	10'-0"	1
102	MAIN	WIC	6'-6 1/8" x 3'-0 1/4"	20	1.8	10'-0"	1
103	MAIN	WC	6'-6 1/8" x 4'-1 3/4"	27	2.5	10'-0"	1
104	MAIN	BATH 2	10'-2 3/4" x 6'-8 7/8"	69	6.4	10'-0"	1
105	MAIN	BUNK ROOM	15'-1 1/4" x 12'-10 5/8"	195	18.1	10'-0"	1
106	MAIN	HALL	20'-6 5/8" x 4'-1 3/8"	84	7.8	10'-0"	1
107	MAIN	SITTING ROOM	11'-3 5/8" x 12'-8 3/4"	144	13.4	10'-0"	1
108	MAIN	BREAKFAST NOOK	15'-2 1/4" x 12'-8 3/4"	193	18.0	10'-0"	1
109	MAIN	HALL	15'-2 1/4" x 5'-3 5/8"	81	7.5	10'-0"	1
110	MAIN	DECK	26'-5 7/8" x 12'-2"	322	30.0	-	1
111	MAIN	GREAT ROOM	26'-5 7/8" x 17'-0 7/8"	452	42.0	12'-0"	1
112	MAIN	OFFICE PORCH	12'-11 7/8" x 11'-3 3/4"	147	13.6	-	1
113	MAIN	MASTER BEDROOM	18'-4 1/4" x 23'-4 1/4"	428	39.8	10'-0"	1
114	MAIN	MASTER PORCH	18'-4 1/4" x 9'-6 1/2"	175	16.3	-	1
115	MAIN	BEDROOM 2	14'-10" x 12'-8"	188	17.5	10'-0"	1
116	MAIN	LAUNDRY	15'-2 3/8" x 9'-1"	138	12.8	10'-0"	1
117	MAIN	MUDROOM	15'-2 3/8" x 5'-1 1/4"	78	7.2	10'-0"	1
119	MAIN	BATH	6'-8 1/8" x 9'-11 1/8"	66	6.2	10'-0"	1
120	MAIN	FLEX ROOM	15'-2 1/4" x 9'-11 1/8"	151	14.0	10'-0"	1
121	MAIN	OFFICE	12'-11 7/8" x 12'-8 3/8"	165	15.3	10'-0"	1
122	MAIN	SIDE PORCH	10'-8 1/4" x 29'-11 1/2"	320	29.8	-	1
123	MAIN	SCULLERY	11'-3 1/8" x 8'-8 1/4"	98	9.1	12'-0"	1
124	MAIN	KITCHEN / DINING	33'-6 1/2" x 23'-2 3/4"	779	72.4	12'-0"	1
125	MAIN	HALL	80'-10 3/8" x 5'-11 3/8"	481	44.7	10'-0"	1
126	MAIN	WIC	10'-6 3/4" x 8'-11 3/8"	95	8.8	10'-0"	1
127	MAIN	WIC	10'-6 3/8" x 8'-11 3/8"	94	8.8	10'-0"	1
128	MAIN	LINEN	6'-2 1/4" x 8'-11 3/8"	55	5.1	10'-0"	1

ROOM SCHEDULE (2 of 2)

NO.	LVL	ROOM	SIZE (C/L)	ft2	m2	CEILING	PHASE
129	MAIN	GARAGE	24'-4 1/8" x 34'-3 1/2"	833	77.4	10'-0"	1
130	MAIN	SIDE ENTRY	11'-3 1/8" x 7'-5 1/8"	84	7.8	-	1
131	MAIN	FOYER	11'-11 3/4" x 25'-9 1/2"	309	28.7	12'-0"	1
132	MAIN	DEN	14'-10" x 29'-6 1/8"	438	40.7	12'-0"	1
133	MAIN	WC	6'-9 3/4" x 3'-10 1/4"	26	2.4	10'-0"	1
134	MAIN	WC	6'-9 3/4" x 3'-9 3/8"	26	2.4	10'-0"	1
135	MAIN	MASTER BATH	21'-7 1/2" x 11'-3"	243	22.6	10'-0"	1
136	MAIN	BREAKFAST	15'-0 1/2" x 6'-3 3/8"	94	8.8	10'-0"	1
137	MAIN	FRONT PORCH	52'-2" x 15'-3 1/8"	796	74.0	-	1
B01	BSMT	MUSIC ROOM	20'-5 7/8" x 12'-9 3/4"	262	24.4	12'-0"	2 (FINISH)
B03	BSMT	GYM	11'-5 7/8" x 17'-9 7/8"	205	19.0	12'-0"	2 (FINISH)
B04	BSMT	STORAGE	15'-2 1/2" x 7'-7"	115	10.7	12'-0"	2 (FINISH)
B05	BSMT	UTILITY	15'-2 1/2" x 7'-8 1/4"	117	10.9	12'-0"	2 (FINISH)
B06	BSMT	HALL	5'-3 3/8" x 15'-3 1/4"	81	7.5	12'-0"	2 (FINISH)
B07	BSMT	BATH	6'-10 1/2" x 10'-3 1/8"	70	6.5	12'-0"	2 (FINISH)
B08	BSMT	HOME THEATER	15'-2 7/8" x 21'-2 3/4"	324	30.1	13'-3"	2 (FINISH)
B09	BSMT	POOL EQUIP.	6'-8 1/2" x 12'-9 7/8"	86	8.0	12'-0"	2 (FINISH)
B10	BSMT	HALLWAY	18'-10 1/2" x 1'-1 1/2"	21	2.0	-	2 (FINISH)
B11	BSMT	STAIR UP	4'-7 1/2" x 16'-7 3/8"	77	7.1	12'-0"	1
B12	BSMT	POOL ROOM	26'-11 3/8" x 34'-7 3/8"	933	86.7	12'-0"	2 (FINISH)
B13	BSMT	POOL BATH	6'-8 1/2" x 10'-11 5/8"	74	6.8	12'-0"	2 (FINISH)
B14	BSMT	PET ROOM	15'-2 7/8" x 10'-11 5/8"	167	15.5	12'-0"	2 (FINISH)
B15	BSMT	HALL	48'-10 3/4" x 4'-8 5/8"	231	21.4	12'-0"	2 (FINISH)
B16	BSMT	LIVING ROOM	18'-10 1/2" x 21'-9 3/4"	412	38.2	10'-0"	2 (FINISH)
B17	BSMT	BATH	13'-3 1/8" x 7'-7 5/8"	101	9.4	12'-0"	2 (FINISH)
B18	BSMT	LAUNDRY	14'-3 3/8" x 7'-7 5/8"	109	10.1	12'-0"	2 (FINISH)

SCHEDULE NOTES

- Marks: D/W + 1xx main floor, Bxx basement, numbered in reading order (north to south, west to east).
- Sizes are the model's opening widths (wall centre-line plan); rough openings, frames and hardware are for the supplier. Swing: the direction the leaf opens toward.
- Window U-values per NBC 9.36 for the climate zone (zone 7A/7B: U <= 1.61 W/m2K prescriptive; triple glazing recommended). Safety glazing where NBC 9.6.1.4 requires it (doors, sidelights, near tubs and the pool).
- Room sizes between wall centre lines; areas are gross. Basement rooms are finished in Phase 2.

NEW SINGLE-FAMILY DWELLING

Attached garage, full ICF basement, indoor pool + spa

Rural Alberta - legal land description: TBD

Owner / applicant: _____

NOT FOR CONSTRUCTION

FOR ENGINEERING REVIEW AND PERMIT PREPARATION

Drawn from the OpenSCAD model in github.com/nuniesmith/house (models/) by tools/make_drawings.py.

Dimensions are to wall centre lines unless noted. Walls are drawn 4 3/4" interior, 8 3/8" exterior (2×6 + sheathing + cladding assumed) and 13 1/4" ICF.

Elevations are relative to the main floor = 0'-0". Finished grade is taken at -2'-0" all round until a site survey and grading plan fix it.

Structure, reinforcement, connections and member sizes are by the engineer of record and the truss designer. Verify every dimension before building.

ENGINEER'S REVIEW / SEAL

SHEET TITLE

SCHEDULES - DOORS, WINDOWS, ROOMS

SCALE: NONE

SHEET 14 of 14

A6.0

DATE 2026-11-03

REV. 0 - issued for review